

第五节

隐函数的求导方法

1) 方程在什么条件下才能确定隐函数.

例如, 方程 $x^2 + \sqrt{y} + C = 0$ $\begin{cases} C < 0 \text{ 时, 能确定隐函数} \\ C > 0 \text{ 时, 不能确定隐函数} \end{cases}$

2) 方程能确定隐函数时, 研究其连续性, 可微性及求导方法问题.

本节讨论:

一、一个方程所确定的隐函数 及其导数

二、方程组所确定的隐函数组 及其导数



一、一个方程所确定的隐函数及其导数

定理1. 设函数 $F(x, y)$ 在点 $P(x_0, y_0)$ 的某一邻域内满足

① 具有连续的偏导数;

② $F(x_0, y_0) = 0$;

③ $F_y(x_0, y_0) \neq 0$

则方程 $F(x, y) = 0$ 在点 x_0 的**某邻域内**可唯一确定一个单值连续函数 $y = f(x)$, 满足条件 $y_0 = f(x_0)$, 并有连续导数

$$\frac{dy}{dx} = -\frac{F_x}{F_y} \quad (\text{隐函数求导公式})$$

定理证明从略, 仅就求导公式推导如下:



设 $y = f(x)$ 为方程 $F(x, y) = 0$ 所确定的隐函数, 则

$$F(x, f(x)) \equiv 0$$

↓ 两边对 x 求导

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0$$

↓ 在 (x_0, y_0) 的某邻域内 $F_y \neq 0$

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$



若 $F(x, y)$ 的二阶偏导数也都连续,
则还可求隐函数的二阶导数:

$$\frac{d^2 y}{dx^2} = \frac{\partial}{\partial x} \left(-\frac{F_x}{F_y} \right) + \frac{\partial}{\partial y} \left(-\frac{F_x}{F_y} \right) \cdot \frac{dy}{dx}$$

$$= -\frac{F_{xx}F_y - F_{yx}F_x}{F_y^2} - \frac{F_{xy}F_y - F_{yy}F_x}{F_y^2} \left(-\frac{F_x}{F_y} \right)$$

$$= -\frac{F_{xx}F_y^2 - 2F_{xy}F_xF_y + F_{yy}F_x^2}{F_y^3}$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

$\swarrow \quad \searrow$
 $x \quad y$
 $\quad \quad \downarrow$
 $\quad \quad x$



例1. 验证方程 $\sin y + e^x - xy - 1 = 0$ 在点 $(0,0)$ 某邻域可确定一个单值可导隐函数 $y = f(x)$, 并求

$$\left. \frac{dy}{dx} \right|_{x=0}, \quad \left. \frac{d^2 y}{dx^2} \right|_{x=0}$$

解: 令 $F(x, y) = \sin y + e^x - xy - 1$, 则

① $F_x = e^x - y, F_y = \cos y - x$ 连续;

② $F(0,0) = 0$;

③ $F_y(0,0) = 1 \neq 0$,

由定理1可知, 在 $x = 0$ 的某邻域内方程存在单值可导的隐函数 $y = f(x)$, 且



$$\left. \frac{dy}{dx} \right|_{x=0} = - \left. \frac{F_x}{F_y} \right|_{x=0} = - \left. \frac{e^x - y}{\cos y - x} \right|_{x=0, y=0} = -1$$

$$\left. \frac{d^2 y}{dx^2} \right|_{x=0}$$

$$= - \left. \frac{d}{dx} \left(\frac{e^x - y}{\cos y - x} \right) \right|_{x=0, y=0, y'=-1}$$

$$= - \left. \frac{(e^x - y')(\cos y - x) - (e^x - y)(-\sin y \cdot y' - 1)}{(\cos y - x)^2} \right|_{\begin{matrix} x=0 \\ y=0 \\ y'=-1 \end{matrix}}$$

$$= -3$$



导数的另一求法 — 利用隐函数求导

$$\sin y + e^x - xy - 1 = 0, \quad y = y(x)$$

两边对 x 求导

$$\cos y \cdot y' + e^x - y - xy' = 0 \longrightarrow$$

两边再对 x 求导

$$-\sin y \cdot (y')^2 + \cos y \cdot y'' + e^x - y' - y' - xy'' = 0$$

令 $x = 0$, 注意此时 $y = 0, y' = -1$

$$\left. \frac{d^2 y}{dx^2} \right|_{x=0} = -3$$

$$\begin{aligned} y' \Big|_{x=0} &= -\frac{e^x - y}{\cos y - x} \Big|_{(0,0)} \\ &= -1 \end{aligned}$$



定理2. 若函数 $F(x, y, z)$ 满足:

- ① 在点 $P(x_0, y_0, z_0)$ 的某邻域内具有**连续偏导数**;
- ② $F(x_0, y_0, z_0) = 0$;
- ③ $F_z(x_0, y_0, z_0) \neq 0$,

则方程 $F(x, y, z) = 0$ 在点 (x_0, y_0) 某一邻域内可唯一确定一个单值连续函数 $z = f(x, y)$, 满足 $z_0 = f(x_0, y_0)$, 并有连续偏导数

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

定理证明从略, 仅就求导公式推导如下:



设 $z = f(x, y)$ 是方程 $F(x, y) = 0$ 所确定的隐函数, 则

$$F(x, y, f(x, y)) \equiv 0$$



两边对 x 求偏导

$$F_x + F_z \frac{\partial z}{\partial x} = 0$$



在 (x_0, y_0, z_0) 的某邻域内 $F_z \neq 0$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$$

同样可得

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$



例2. 设 $x^2 + y^2 + z^2 - 4z = 0$, 求 $\frac{\partial^2 z}{\partial x^2}$.

解法1 利用隐函数求导

$$2x + 2z \frac{\partial z}{\partial x} - 4 \frac{\partial z}{\partial x} = 0 \longrightarrow \frac{\partial z}{\partial x} = \frac{x}{2-z}$$

再对 x 求导

$$2 + 2\left(\frac{\partial z}{\partial x}\right)^2 + 2z \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x^2} = 0$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{1 + \left(\frac{\partial z}{\partial x}\right)^2}{2-z} = \frac{(2-z)^2 + x^2}{(2-z)^3}$$



解法2 利用公式

设 $F(x, y, z) = x^2 + y^2 + z^2 - 4z$

则 $F_x = 2x, F_z = 2z - 4$

$$\therefore \frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{x}{z-2} = \frac{x}{2-z}$$

两边对 x 求偏导

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{x}{2-z} \right) = \frac{(2-z) + x \frac{\partial z}{\partial x}}{(2-z)^2} = \frac{(2-z)^2 + x^2}{(2-z)^3}$$



例3. 设 $F(x, y)$ 具有连续偏导数, 已知方程 $F(\frac{x}{z}, \frac{y}{z}) = 0$, 求 dz .

解法1 利用偏导数公式. 设 $z = f(x, y)$ 是由方程 $F(\frac{x}{z}, \frac{y}{z}) = 0$ 确定的隐函数, 则

$$\frac{\partial z}{\partial x} = - \frac{F'_1 \cdot \frac{1}{z}}{F'_1 \cdot (-\frac{x}{z^2}) + F'_2 \cdot (-\frac{y}{z^2})} = \frac{z F'_1}{x F'_1 + y F'_2}$$

$$\frac{\partial z}{\partial y} = - \frac{F'_2 \cdot \frac{1}{z}}{F'_1 \cdot (-\frac{x}{z^2}) + F'_2 \cdot (-\frac{y}{z^2})} = \frac{z F'_2}{x F'_1 + y F'_2}$$

故 $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = \frac{z}{x F'_1 + y F'_2} (F'_1 dx + F'_2 dy)$



解法2 微分法. 对方程两边求微分:

$$F\left(\frac{x}{z}, \frac{y}{z}\right) = 0$$

$$F_1' \cdot d\left(\frac{x}{z}\right) + F_2' \cdot d\left(\frac{y}{z}\right) = 0$$

$$F_1' \cdot \left(\frac{zdx - xdz}{z^2}\right) + F_2' \cdot \left(\frac{zdy - ydz}{z^2}\right) = 0$$

$$\frac{x F_1' + y F_2'}{z^2} dz = \frac{F_1' dx + F_2' dy}{z}$$

$$dz = \frac{z}{x F_1' + y F_2'} (F_1' dx + F_2' dy)$$



二、方程组所确定的隐函数组及其导数

隐函数存在定理还可以推广到方程组的情形.

以两个方程确定两个隐函数的情况为例, 即

$$\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases} \longrightarrow \begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$$

由 F 、 G 的偏导数组成的行列式

$$J = \frac{\partial(F, G)}{\partial(u, v)} = \begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}$$

称为 F 、 G 的雅可比行列式.



定理3. 设函数 $F(x, y, u, v), G(x, y, u, v)$ 满足:

① 在点 $P(x_0, y_0, u_0, v_0)$ 的某一邻域内具有连续偏导数;

② $F(x_0, y_0, u_0, v_0) = 0, G(x_0, y_0, u_0, v_0) = 0$;

③ $J \bigg|_P = \frac{\partial(F, G)}{\partial(u, v)} \bigg|_P \neq 0,$

则方程组 $F(x, y, u, v) = 0, G(x, y, u, v) = 0$ 在点 (x_0, y_0) 的某一邻域内可**唯一**确定一组满足条件 $u_0 = u(x_0, y_0), v_0 = v(x_0, y_0)$ 的**单值连续函数** $u = u(x, y), v = v(x, y)$, 且有偏导数公式:



$$\frac{\partial u}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(\underline{x}, v)} = -\frac{1}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}} \begin{vmatrix} F_x & F_v \\ G_x & G_v \end{vmatrix} \quad (\text{P85})$$

$$\frac{\partial u}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(\underline{y}, v)} = -\frac{1}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}} \begin{vmatrix} F_y & F_v \\ G_y & G_v \end{vmatrix}$$

$$\frac{\partial v}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, \underline{x})} = -\frac{1}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}} \begin{vmatrix} F_u & F_x \\ G_u & G_x \end{vmatrix}$$

定理证明略.
仅推导偏导
数公式如下:

$$\frac{\partial v}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, \underline{y})} = -\frac{1}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}} \begin{vmatrix} F_u & F_y \\ G_u & G_y \end{vmatrix}$$



设方程组 $\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$ 有隐函数组 $\begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$, 则

$$\begin{cases} F(x, y, u(x, y), v(x, y)) \equiv 0 \\ G(x, y, u(x, y), v(x, y)) \equiv 0 \end{cases}$$

两边对 x 求导得 $\begin{cases} F_x + F_u \cdot \frac{\partial u}{\partial x} + F_v \cdot \frac{\partial v}{\partial x} = 0 \\ G_x + G_u \cdot \frac{\partial u}{\partial x} + G_v \cdot \frac{\partial v}{\partial x} = 0 \end{cases}$

这是关于 $\frac{\partial u}{\partial x}$, $\frac{\partial v}{\partial x}$ 的线性方程组, 在点 P 的某邻域内

系数行列式 $J = \begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix} \neq 0$, 故得 $\frac{\partial u}{\partial x} = - \frac{\begin{vmatrix} F_x & F_v \\ G_x & G_v \end{vmatrix}}{\begin{vmatrix} F_u & F_v \\ G_u & G_v \end{vmatrix}}$



$$\frac{\partial u}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, v)}$$

$$\frac{\partial v}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, x)}$$

同样可得

$$\frac{\partial u}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, v)}$$

$$\frac{\partial v}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, y)}$$



例4. 设 $xu - yv = 0$, $yu + xv = 1$, 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

解: 方程组两边对 x 求导, 并移项得

$$\begin{cases} x \frac{\partial u}{\partial x} - y \frac{\partial v}{\partial x} = -u \\ y \frac{\partial u}{\partial x} + x \frac{\partial v}{\partial x} = -v \end{cases}$$

由题设 $J = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2 + y^2 \neq 0$

故有 $\begin{cases} \frac{\partial u}{\partial x} = \frac{1}{J} \begin{vmatrix} -u & -y \\ -v & x \end{vmatrix} = -\frac{xu + yv}{x^2 + y^2} \\ \frac{\partial v}{\partial x} = \frac{1}{J} \begin{vmatrix} x & -u \\ y & -v \end{vmatrix} = -\frac{xv - yu}{x^2 + y^2} \end{cases}$

练习: 求 $\frac{\partial u}{\partial y}, \frac{\partial v}{\partial y}$

答案:

$$\begin{cases} \frac{\partial u}{\partial y} = -\frac{yu - xv}{x^2 + y^2} \\ \frac{\partial v}{\partial y} = -\frac{xu + yv}{x^2 + y^2} \end{cases}$$



例5. 设函数 $x = x(u, v)$, $y = y(u, v)$ 在点 (u, v) 的某一邻域内有连续的偏导数, 且 $\frac{\partial(x, y)}{\partial(u, v)} \neq 0$

1) 证明函数组 $\begin{cases} x = x(u, v) \\ y = y(u, v) \end{cases}$ 在与点 (u, v) 对应的点 (x, y) 的某一邻域内唯一确定一组单值、连续且具有连续偏导数的反函数 $u = u(x, y)$, $v = v(x, y)$.

2) 求 $u = u(x, y)$, $v = v(x, y)$ 对 x, y 的偏导数.

解: 1) 令 $F(x, y, u, v) \equiv x - x(u, v) = 0$

$$G(x, y, u, v) \equiv y - y(u, v) = 0$$



则有
$$J = \frac{\partial(F, G)}{\partial(u, v)} = \frac{\partial(x, y)}{\partial(u, v)} \neq 0,$$

由定理 3 可知结论 1) 成立.

2) 求反函数的偏导数.

$$\begin{cases} x \equiv x(u(x, y), v(x, y)) \\ y \equiv y(u(x, y), v(x, y)) \end{cases} \quad (1)$$

①式两边对 x 求导, 得

$$\begin{cases} 1 = \frac{\partial x}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial x}{\partial v} \cdot \frac{\partial v}{\partial x} \\ 0 = \frac{\partial y}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial y}{\partial v} \cdot \frac{\partial v}{\partial x} \end{cases} \quad (2)$$



注意 $J \neq 0$, 从方程组②解得

$$\frac{\partial u}{\partial x} = \frac{1}{J} \begin{vmatrix} 1 & \frac{\partial x}{\partial v} \\ 0 & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{1}{J} \frac{\partial y}{\partial v},$$

$$\frac{\partial v}{\partial x} = \frac{1}{J} \begin{vmatrix} \frac{\partial x}{\partial u} & 1 \\ \frac{\partial y}{\partial u} & 0 \end{vmatrix} = -\frac{1}{J} \frac{\partial y}{\partial u}$$

同理, ①式两边对 y 求导, 可得

$$\frac{\partial u}{\partial y} = -\frac{1}{J} \frac{\partial x}{\partial v},$$

$$\frac{\partial v}{\partial y} = \frac{1}{J} \frac{\partial x}{\partial u}$$



例5的应用: 计算极坐标变换 $x = \overset{u}{r} \cos \overset{v}{\theta}$, $y = r \sin \theta$

的反变换的导数.

由于 $J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$

$$\begin{aligned} \frac{\partial r}{\partial x} &= \frac{1}{J} \frac{\partial y}{\partial \theta} \\ \frac{\partial \theta}{\partial x} &= -\frac{1}{J} \frac{\partial y}{\partial r} \end{aligned}$$

所以 $\frac{\partial r}{\partial x} = \frac{1}{J} \frac{\partial y}{\partial \theta} = \frac{1}{r} r \cos \theta = \cos \theta = \frac{x}{\sqrt{x^2 + y^2}}$

$$\frac{\partial \theta}{\partial x} = -\frac{1}{J} \frac{\partial y}{\partial r} = -\frac{1}{r} \sin \theta = -\frac{y}{x^2 + y^2}$$

同样有 $\frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$ $\frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2}$



内容小结

1. 隐函数(组) 存在定理
2. 隐函数 (组) 求导方法

方法1. 利用复合函数求导法则直接计算 ;

方法2. 利用微分形式不变性 ;

方法3. 代公式 .

思考与练习

设 $z = f(x + y + z, xyz)$, 求 $\frac{\partial z}{\partial x}$, $\frac{\partial x}{\partial z}$, $\frac{\partial x}{\partial y}$.



提示: $z = f(x + y + z, xyz)$

$$\bullet \quad \frac{\partial z}{\partial x} = f'_1 \cdot \left(1 + \frac{\partial z}{\partial x}\right) + f'_2 \cdot \left(yz + xy \frac{\partial z}{\partial x}\right)$$

$$\implies \frac{\partial z}{\partial x} = \frac{f'_1 + yzf'_2}{1 - f'_1 - xyf'_2}$$

$$\bullet \quad 1 = f'_1 \cdot \left(\frac{\partial x}{\partial z} + 1\right) + f'_2 \cdot \left(yz \frac{\partial x}{\partial z} + xy\right)$$

$$\implies \frac{\partial x}{\partial z} = \frac{1 - f'_1 - xyf'_2}{f'_1 + yzf'_2}$$

$$\bullet \quad 0 = f'_1 \cdot \left(\frac{\partial x}{\partial y} + 1\right) + f'_2 \cdot \left(yz \frac{\partial x}{\partial y} + xz\right)$$

$$\implies \frac{\partial x}{\partial y} = -\frac{f'_1 + xzf'_2}{f'_1 + yzf'_2}$$



解法2. 利用全微分形式不变性同时求出各偏导数.

$$z = f(x + y + z, xyz)$$

$$dz = f'_1 \cdot (dx + dy + dz) + f'_2 \cdot (yz dx + xz dy + xy dz)$$

解出 dx :

$$dx = \frac{-(f'_1 + xzf'_2)dy + (1 - f'_1 - xyf'_2)dz}{f'_1 + yzf'_2}$$

由 dy, dz 的系数即可得 $\frac{\partial x}{\partial y}, \frac{\partial x}{\partial z}$.

作业

P87 3, 6, 7, *9, 10(1); (3), 11

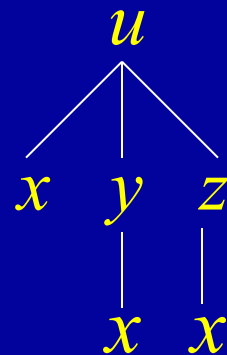


备用题 1. 设 $u = f(x, y, z)$ 有连续的一阶偏导数，
又函数 $y = y(x)$ 及 $z = z(x)$ 分别由下列两式确定：

$e^{xy} - xy = 2$, $e^x = \int_0^{x-z} \frac{\sin t}{t} dt$, 求 $\frac{du}{dx}$. (2001 考研)

解： 两个隐函数方程两边对 x 求导，得

$$\begin{cases} e^{xy}(y + xy') - (y + xy') = 0 \\ e^x = \frac{\sin(x-z)}{x-z} (1 - z') \end{cases}$$



解得 $y' = -\frac{y}{x}, \quad z' = 1 - \frac{e^x(x-z)}{\sin(x-z)}$

因此 $\frac{du}{dx} = f'_1 - \frac{y}{x} f'_2 + \left[1 - \frac{e^x(x-z)}{\sin(x-z)} \right] f'_3$



2. 设 $y = y(x)$, $z = z(x)$ 是由方程 $z = x f(x + y)$ 和 $F(x, y, z) = 0$ 所确定的函数, 求 $\frac{dz}{dx}$. (1999 考研)

解法1 分别在各方程两端对 x 求导, 得

$$\begin{cases} z' = f + x \cdot f' \cdot (1 + y') \\ F_x + F_y \cdot y' + F_z \cdot z' = 0 \end{cases} \Rightarrow \begin{cases} -x f' \cdot y' + \underline{z'} = f + x f' \\ F_y \cdot y' + F_z \cdot \underline{z'} = -F_x \end{cases}$$

$$\therefore \frac{dz}{dx} = \frac{\begin{vmatrix} -x f' & f + x f' \\ F_y & -F_x \end{vmatrix}}{\begin{vmatrix} -x f' & 1 \\ F_y & F_z \end{vmatrix}} = \frac{(f + x f') F_y - x f' \cdot F_x}{F_y + x f' \cdot F_z} \quad (F_y + x f' \cdot F_z \neq 0)$$



解法2 微分法.

$$z = xf(x+y), \quad F(x, y, z) = 0$$

对各方程两边分别求微分:

$$\begin{cases} dz = f dx + xf' \cdot (dx + dy) \\ F_1' dx + F_2' dy + F_3' dz = 0 \end{cases}$$

化简得

$$\begin{cases} (f + xf') dx + xf' dy - dz = 0 \\ F_1' dx + F_2' dy + F_3' dz = 0 \end{cases}$$

消去 dy 可得 $\frac{dz}{dx}$.



二元线性代数方程组解的公式

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

解: $x = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$

$$y = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$



雅可比(1804 – 1851)

德国数学家. 他在数学方面最主要的成就是和挪威数学家阿贝儿相互独立地奠定了椭圆函数论的基础. 他对行列式理论也作了奠基性的工作. 在偏微分方程的研究中引进了“雅可比行列式”并应用在微积分中. 他的工作还包括代数学, 变分法, 复变函数和微分方程, 在分析力学, 动力学及数学物理方面也有贡献. 他在柯尼斯堡大学任教18年, 形成了以他为首的学派.



雅可比, C. G. J.

